AI-Based Rainwater Harvesting Recommendation System (Bengaluru)

A data-driven tool that recommends a suitable rainwater harvesting (RWH) option for a location in Bengaluru, based on map, rainfall, physical, demographic and household-pattern data.

> **Status:** Early stage. Problem scope, data sources and methodology are defined; model development is in progress.

Problem

Bengaluru receives most of its rainfall in a few monsoon months, and much of it is lost as runoff in paved urban areas. Households usually pick RWH structures using generic rules or vendor advice, without checking local rainfall, soil, roof area or water demand. This project aims to give location-specific recommendations instead.

What it does

Given an address or map pin plus a few household details, the system will:

1. Estimate harvestable rainwater for the location
2. Recommend a structure type (storage tank, recharge pit, borewell recharge, or a combination)
3. Suggest an approximate size and cost
4. Compare expected supply with household demand and show the likely water saving

Inputs

Category	Examples
Map / location	Pin or address, ward or hobli, OpenStreetMap layers
Rainfall	Monthly and annual rainfall by taluk/hobli
Physical	Roof area and type, open area, soil type, groundwater depth, elevation/slope
Demographic	Household size, ward-level population density
Household pattern	House type (independent, apartment, villa), daily water use,

Category	Examples
	water source (borewell, Cauvery supply)

Approach

```
Input data -> Preprocessing ->
Recommendation model -> Harvest potential
estimate
   -> Structure recommendation -> Sizing
and cost -> Demand-supply comparison
   -> Water saving and recharge analysis ->
Feedback to improve model
```

Core estimate:

```
Harvest (litres) = Rainfall (mm) x Roof
area (m^2) x Runoff coefficient
```

A rule-based calculator comes first. A machine learning layer (for example a decision tree or random forest) then classifies the best structure type from site and household features.

Data sources

- **Rainfall:** KSNDMC taluk and hobli rainfall records (published via OpenCity); IMD Bengaluru daily

rainfall reports

- **Roof area:** Building footprints (Google Open Buildings or Microsoft Building Footprints)
- **Map layers:** OpenStreetMap
- **Ward boundaries and population:** OpenCity ward data, Census 2011
- **Groundwater:** India-WRIS, CGWB
- **Soil:** NBSS&LUP, SoilGrids
- **Elevation:** NASA SRTM

Please check each source's licence and availability before use. Household water-use figures will start as standard assumptions that users can override.

Planned tech stack

- Python (FastAPI backend, pandas, scikit-learn)
- Streamlit or React frontend with Leaflet for the map
- CSV or PostGIS for data storage

Roadmap

- [ ] Load and clean KSNDMC rainfall data by hobli
- [ ] Rainfall and harvest-potential calculator
- [ ] Map input and zone lookup

- [ ] Recommendation model (structure type and size)
- [ ] Cost estimate and demand-supply comparison
- [ ] Web UI and result report
- [ ] Validation against standard RWH design guidelines

Limitations

- Covers Bengaluru only for now
- Recommendations are estimates, not engineering designs; a site visit and professional design are still needed
- Household usage patterns are assumed unless the user provides them

Related work

The approach builds on GIS and multi-criteria (AHP) studies of RWH site suitability, ML-based RWH potential mapping, rooftop potential studies, and Bengaluru groundwater research (Sekhar et al., *Sustainability*, 2018).

Sustainability goals

Supports SDG 6 (Clean Water and Sanitation), SDG 11 (Sustainable Cities and Communities) and SDG 13

(Climate Action).

Course context

Environmental Science project (BAIP508), Department of AI & ML, BMS Institute of Technology & Management.

License

To be decided.